GAURAV KUMAR MAURYA

Computer Science Engineering | AI & Android Developer

gauravmauryagkm55@gmail.com | +91 9335176236 | Varanasi, Uttar Pradesh, India

LinkedIn | GitHub | LeetCode | Portfolio

PROFESSIONAL SUMMARY

Computer Science Engineering graduate with deep technical expertise in Android application development using Kotlin and Android Studio. Hands-on experience building end-to-end data pipelines and implementing on-device edge AI and machine learning models. Proven track record of engineering complex mobile systems — including an enterprise-grade real-time road safety vision platform alongside data-driven ML projects. Well-versed in Python-based data science (Pandas, NumPy, Scikit-Learn) and foundational Generative AI architectures. Seeking a technical entry-level role to build robust data, AI, and mobile solutions at scale.

TECHNICAL SKILLS

Programming Languages: Python, Kotlin, Java, SQL

AI & Data Science: TensorFlow Lite (TFLite), OpenCV, Object Detection, INT8 Model Quantization, Machine Learning Pipelines, Exploratory Data Analysis (EDA)

Mobile & Frontend: Android Studio, CameraX API, Jetpack Compose, Material Design 3, HTML, CSS

Databases & Cloud: Room Database (SQLite/WAL), Firebase Realtime Database, Firebase Analytics & Crashlytics

Advanced APIs: Google ARCore (AR), Wear OS Smartwatch Data Layer, Ultra-Wideband (UWB) Ranging,FusedLocationProviderClient (GPS)

Dev Tools: Git, GitHub, Android Studio Profiler, Gradle (Kotlin DSL), Figma

PROJECTS

SafeStrideAI: AI-Based Real-Time Road Safety App — *Jan 2026 – May 2026*

Major College Project | Kotlin · Android Studio · TensorFlow Lite · OpenCV · Google ARCore · Room DB · WearOS · UWB

- **Edge AI & Model Optimization:** Integrated a TFLite object detection model tracking pedestrians and vehicles on-device at 20–30 FPS without internet or external server dependency.
- **INT8 Quantization:** Applied post-training integer quantization to compress the deep learning model from 22 MB to 4.18 MB, cutting inference latency by 55–70%.
- **Computer Vision Pipeline:** Implemented Lucas-Kanade Sparse Optical Flow via OpenCV to compute per-frame motion vectors for velocity and heading calculations across detection bounding boxes.
- **Predictive Risk Analytics:** Built a Time-to-Collision (TTC) engine evaluating convergence trajectories, achieving a verified 91.7% danger-alert accuracy profile.
- **Multi-Thread & Storage Architecture:** Designed a three-thread layout (UI, Inference, DB) with asynchronous SQLite Room DB writes using Write-Ahead Logging (WAL) for zero data corruption.
- **Multi-Sensory Alert System:** Deployed Android Foreground Services syncing collision warnings to Google ARCore overlays and Wear OS smartwatches via the Wearable Data Layer API in under 200 ms.

Credit Card Default Prediction System — *Aug 2025 – Oct 2025*

Python · Pandas · NumPy · Scikit-Learn

- **Supervised ML:** Built a complete automated pipeline predicting credit card defaults across 30,000 customer records.
- **EDA:** Performed data wrangling, missing value handling, and feature engineering via Pandas and NumPy, identifying critical correlations between high credit utilization and late payment patterns.
- **Model Selection:** Trained and cross-evaluated multiple Scikit-Learn classifiers, comparing accuracy vs. recall metrics to select the optimal deployment model.

30-Day Fitness Android Application — *Jul 2024 – Sep 2024*

Kotlin · Android Studio · SQLite · Material Design · Firebase

- Architected a standalone, cross-device workout scheduler with multi-screen navigation, day-by-day progression tracking, and Material Design 3 UI.
- Combined SQLite local persistent storage with Firebase Realtime Database for seamless cross-session data synchronization.

PROFESSIONAL EXPERIENCE

Android App Development Intern *Jul 2024 – Sep 2024*

Internshala — *Remote*

- **Feature Engineering:** Delivered clean, production-ready Kotlin feature modules for a distributed mobile application following agile project roadmaps.
- **Agile Collaboration:** Synchronized with product managers, UX designers, and QA testers across sprint cycles, ensuring on-time delivery of key feature modules.
- **Performance Tuning:** Diagnosed memory leaks and layout bottlenecks using Android Studio Profiler, reducing frame drops by 45% and improving overall app stability.

EDUCATION

B.Tech – Computer Science & Engineering *2022 – 2026*

Ch. Devi Lal State Institute of Engineering and Technology (SIET), Haryana | Score: **CGPA: 8.81**

Class 12th – UP Board *2021*

Govt. Queen's Inter College, Varanasi | Score: **76.8%**

Class 10th – UP Board *2019*

Kisan Inter College, Barahani, Chandauli | Score: **76.0%**

CERTIFICATIONS

- **Fundamentals of AI and Generative AI** – Oracle *(2025)*
- **Data Science & Machine Learning** – YBI Foundation *(2025)*
- **Python 101 for Data Science** – IBM *(2024)*
- **Android App Development Course** – Internshala *(2024)*